

1. A network shows an open circuit voltage of 9.0 V and a short-circuit current of 1.5 A at its terminals. Find R_{th} . In & the terminal voltage for a $4\text{-}\Omega$ load.

$\Rightarrow$ Open-circuit voltage, $V_{th} = 9\text{ V}$

Short-circuit current, $I_N = I_{sc} = 1.5\text{ A}$

Load Resistance, $R_L = 4\text{-}\Omega$

To find R_{th} ,

From Thevenin - Norton Relationships, $R_{th} = \frac{V_{th}}{I_N} = \frac{9}{1.5} = 6\text{-}\Omega$

$\therefore R_N = R_{th} = 6\text{-}\Omega$

Terminal voltage for $4\text{-}\Omega$ load,

Using Thevenin equivalent circuit,

$$V_L = \frac{V_{th} R_L}{R_{th} + R_L} = 9 \left(\frac{4}{6 + 4} \right) = 3.6\text{ V}$$

2. A 100 mH inductor & a $47\text{ }\mu\text{F}$ capacitor are placed in series. Find the frequency at which their reactances are equal.

$\Rightarrow$ Given,

$$L = 100\text{ mH} = 0.1\text{ H}$$

$$C = 47\text{ }\mu\text{F} = 47 \times 10^{-6}\text{ F}$$

At equal reactance, $X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C}$

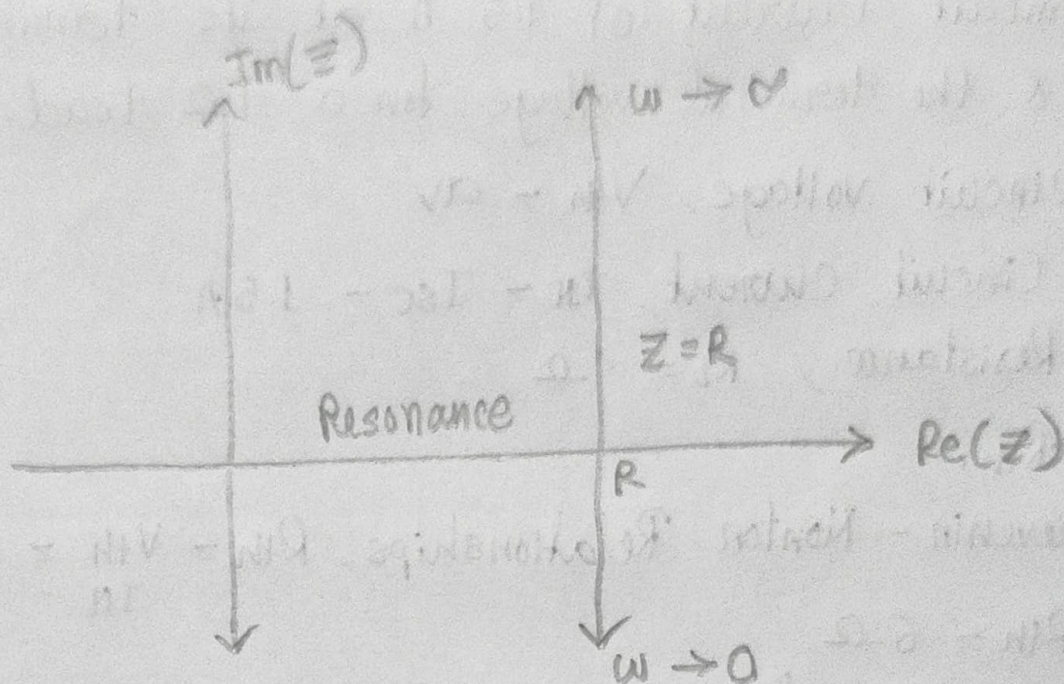
$$\therefore \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{(0.1)(47 \times 10^{-6})}} = 461.9\text{ rad/s}$$

$$\text{Now, } f_0 = \frac{\omega_0}{2\pi} = \frac{461.9}{2\pi} = 73.5\text{ Hz.}$$

Impedance locus,

The locus is a vertical straight line at $\text{Re}(z) = R$:



$$Z = \left(\frac{R}{1 + \omega^2 LC^2} \right) = \frac{R}{1 + \omega^2 LC^2}$$

3. An inductor voltage waveform is $+20V$ for $3ms$ & $-V_2$ for the remaining $2ms$ of a $5ms$ Period. Find V_2 .

⇒ Given,

Total Period = $5ms$.

At steady state operations,

$$\int_0^T v_L(t) dt = 0$$

∴ Positive & negative volt-seconds must be equal.

Apply volt-second balance,

$$(20)(3 \times 10^{-3}) + (-V_2)(2 \times 10^{-3}) = 0$$

$$\Rightarrow 60 \times 10^{-3} - 2V_2 \times 10^{-3} = 0$$

$$\Rightarrow 60 = 2V_2 \Rightarrow V_2 = 30V$$

∴ waveform of Inductor $+20V$ for $3ms$ & $-30V$ for $2ms$.

4. Sketch the impedance locus of a series RLC circuit as ω is swept from 0 to ∞ , & identify the sketch at which the locus crosses the real axis.

⇒ For a series RLC circuit,

$$Z = R + jX_L - jX_C$$

$$\therefore X_L = \omega L \text{ \& } X_C = \frac{1}{\omega C}$$

$$\therefore Z = R + j(\omega L - 1/\omega C)$$

$$\therefore \operatorname{Re}(Z) = R \text{ \& } \operatorname{Im}(Z) = \omega L - 1/\omega C$$

At $\omega = 0$, $X_L = 0$ but $X_C \rightarrow \infty$ ∴ $\operatorname{Im}(Z) \rightarrow -\infty$
so, the locus starts at $R - j\infty$

At resonance,

The locus crosses the real axis when $X_L = X_C$

$$\therefore \omega L = 1/\omega C \Rightarrow \omega_0 = 1/\sqrt{LC}$$

At this point, $X_L - X_C = 0 \Rightarrow$ so, $Z = R$

At $\omega \rightarrow \infty$, $X_L \rightarrow \infty$ & $X_C \rightarrow 0$ ∴ $\operatorname{Im}(Z) \Rightarrow +\infty$

So, the locus goes toward: $R + j\infty$

5. Using Euler's Formula, express $v(t) = 20 \cos(314t + 30^\circ)$ as a Phasor, & state the frequency in Hz.

$\Rightarrow$ Given,

$$v(t) = 20 \cos(314t + 30^\circ)$$

standard sinusoidal form, $v(t) = v_m \cos(\omega t + \phi)$

Here, $v_m = 20$, $\omega = 314 \text{ rad/s}$, $\phi = 30^\circ$

$\therefore$ Phasor is $20 \angle 30^\circ \text{ V} = V'$

$$V_{\text{rms}} = \frac{20}{\sqrt{2}} = 14.14 \text{ V} \text{ then, } V' = 14.14 \angle 30^\circ \text{ V}_{\text{rms}}$$

we know,

$$\omega = 2\pi f \Rightarrow f = 314 / 2\pi \Rightarrow 49.97 \text{ Hz} = f \Rightarrow \approx 50 \text{ Hz}.$$